NABILAH NUR AISYAH N. - 2510514020 - A

No.

Date :

1.) $\int (2-3x)^2 dx$

$\int 4 - 12x + 9x^2 \, dx$

$\int \frac{9x^{2+1}}{2+1} - \frac{12x^{1+1}}{1+1} + \frac{4x^{0+1}}{0+1}$

$\int 3x^3 - 6x^2 + 4x$

2.) $\int 2 \sin x \, dx$

$= \int 2(-\cos x) + C$

$\int -2\cos x + C$

$= -2\cos x + C$

3.) $\int (1 + \sqrt[3]{x}) \, dx$

$\int x^{\frac{1}{3}} + 1 \, dx$

$\int \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1} + \frac{1x^{0+1}}{0+1} \, dx$

$\int \frac{x^{\frac{4}{3}}}{4/3} + x \, dx$

$\cancel{\int \sqrt[3]{\frac{3}{4}x^4} + x} = \frac{3}{4}\sqrt[3]{x^4} + x + C$

$\cancel{= \sqrt[3]{\frac{3}{4}x^4} + x + C} = \frac{3}{4}x\sqrt[3]{x} + x + C$

4.) $\int \frac{2x-1}{x^2} \, dx$

$\int 2x^{1-2} - 1x^{0-2} \, dx$

$\int 2x^{-1} - x^{-2} \, dx$

$= \frac{2x^{-1+1}}{-1+1} - \frac{x^{-2+1}}{-2+1} \, dx$

$= 2\ln|x| - \frac{x^{-1}}{-1} \, dx$

$= 2\ln|x| + \frac{1}{x} \, dx + C$

5.) $\int x\sqrt{x} \, dx$

$\int x^1 \cdot x^{\frac{1}{2}} \, dx$

$\int x^{\frac{3}{2}} \, dx$

$= \frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1}$

$= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + C$

$= \frac{2\sqrt{x^5}}{5} + C$

$= \frac{2x^2\sqrt{x}}{5} + C$

6.) $\int \frac{2}{x^3} - \frac{1}{x^2} \, dx$

$\int 2x^{-3} - x^{-2} \, dx$

$= \frac{2x^{-2}}{-2} - \frac{x^{-1}}{-1} \, dx + C$

$= -x^{-2} + x^{-1} + C$

$= -\frac{1}{x^2} + \frac{1}{x} + C$

7.) $\int (\cos x + \sin 2x) \, dx$

$\int \sin x + -\frac{1}{2}\cos 2x$

$= \sin x - \frac{1}{2}\cos 2x$

8.) $\int \cos^2 x \, dx$

$\int \frac{1 + \cos 2x}{2} \, dx$

$\frac{1}{2}\int 1 + \cos 2x \, dx$

$\frac{1}{2}\int x + \frac{1}{2}\sin 2x$

$= \frac{1}{2}x + \frac{1}{4}\sin 2x + C$

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(9) $\int 3x(x+1)\,dx$

$\int 3x^2 + 3x \; dx$

$= \frac{3x^3}{3} + \frac{3x^2}{2} + C$

$= x^3 + \frac{3}{2}x^2 + C$

(10) $\int \frac{dx}{3\sqrt{x^5}} \; dx$

$= \int \frac{1}{3\sqrt{x^5}} \; dx$

$= \int \frac{1}{3/-\frac{3}{2}} x^{-\frac{5}{2}+\frac{2}{2}} dx$

$= -\frac{2}{9} x^{-\frac{5}{2}+\frac{2}{2}} + C$

$= -\frac{2}{9} x^{-\frac{3}{2}} + C$

$= -\frac{2}{9\sqrt{x^{-3}}} + C$

(11) $\int \frac{1}{3}x^3 - 6x^2 + 8x \; dx$

$= \frac{\frac{1}{3}}{4} x^4 - \frac{6x^3}{3} + \frac{8x^2}{2} + C$

$= \frac{1}{12}x^4 - 2x^3 + 4x^2 + C$

(12) $\int \frac{1}{x} + 2x^2$

$= \int x^{-1} + 2x^2$

$= \frac{x^{-1+1}}{-1+1} + \frac{2x^3}{3} + C$

$= \ln|x| + \frac{2}{3}x^3 + C$

(13) $\int 2x - 3 \; dx$

$= \frac{2x^{1+1}}{2} - 3x$

$f(x) = x^2 - 3x + C$

$f(1) = 1 - 3 + C = -6$

$= -2 + C = -6$

$C = 9$

$f(x) = x^2 - 3x + 9$

(14) $\int_{-2}^{2} (4x + a)\,dx = 12$

$= \left[\frac{4x^2}{2} + ax\right]_{-2}^{2} = 12$

$= \left[2x^2 + ax\right]_{-2}^{2} = 12$

$2(2)^2 + a(2) - (2(-2)^2 + a(-2)) = 12$

$8 + 2a - 8 + 2a = 12$

$4a = 12$

$a = 3$

(15) $\int x^2 - 16 \; dx$

$= \frac{1}{3}x^3 - 16x$

$x = 4$

$\frac{1}{3}64 - 16 \cdot 4$

$=$

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(15.) $\int_0^4 (x^2-16)\,dx$

$\int_0^4 \left(\frac{x^{2+1}}{3} - 16x\right) + c$

$= \int_0^4 \left(\frac{x^3}{3} - 16\right)dx$

$= \left[16x - \frac{x^3}{3}\right]_0^4$

$= \left(64 - \frac{64}{3}\right) - 0$

$= \frac{192-64}{3} = \frac{128}{3}$

$x = 4$ dibawah sb x

$0 - F(x)$

(16.) $\int \left(1 + \frac{1}{x^2}\right) dx$

$\int (x + x^{-2})\,dx$

$\int (x^{1+1} + x^{-2+1})\,dx$

$= (x^2 + x^{-1}) + c$

$= x^2 + \frac{1}{x} + c$

(17) $V = \pi \int_a^b y^2\,dx$

$y = 8 - 2x$

$x = 1 \rightarrow x = 3$

$V = \pi \int_1^3 (8-2x)^2\,dx$

$(8-2x)^2 = 64 - 32x + 4x^2)$

$V = \pi \int_1^3 (64 - 32x + 4x^2)\,dx$

$= \pi \left[64x - 16x^2 + \frac{4}{3}x^3\right]_1^3$

$= \pi \left(84 - \frac{140}{3}\right)$

$= \pi \cdot \frac{104}{3}$

$= \frac{104}{3}\pi$

(18.) $\int_{-1}^{1} (1 - x^2)\,dx$

$= \left[-\frac{x^3}{3} + x\right]_{-1}^{1}$

$= \left(-\frac{1}{3} + 1\right) - \left(-\frac{1}{3} - 1\right)$

$= -\frac{1}{3} + 1 + \frac{1}{3} + 1$

$= 2$

(19.) $\int_0^4 \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx$

$= \left[\frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}}\right]_0^4$

$= \left[\frac{2}{3}x^{\frac{3}{2}} + 2\sqrt{x}\right]_0^4$

$= \left[\frac{2}{3}\sqrt{x^3} + 2\sqrt{x}\right]_0^4$

$= \frac{2}{3}\, 4 \cdot 2 + 2 \cdot 2$

$= \frac{2}{3} \cdot 8 + 2 \cdot 2$

$= \frac{16}{3} + \frac{12}{3} = \frac{28}{3}$

(20.) $\int_{-2}^{0} (2 - x)\,dx$

$= \left[2x - \frac{x^2}{2}\right]_{-2}^{0}$

$= 0 - (-4 - 2)$

$= 0 - (-6)$

$= 6$

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(21.) $P = ? \mid P > 0$

$\int_0^P x(1-x)\,dx = 0$

$\int_0^P x - x^2\,dx = 0$

$\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^P = 0$

$\frac{P^2}{2} - \frac{P^3}{3} = 0$

$P^2\left(\frac{1}{2} - \frac{P}{3}\right) = 0$

$\frac{1}{2} - \frac{P}{3} = 0$

$\frac{3 - 2P}{6} = 0$

$2P = 3$

$P = \frac{3}{2}$ //

(22) apakah

$\int_1^3 9x^3\,dx + \int_3^9 9x^3\,dx = \int_1^9 9x^3\,dx$

benar ?

→ sifat :

$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$

pembuktian : (kiri)

$\int_1^3 9x^3\,dx = [x^9]_1^3 = 81 - 1 = 80$

$\int_3^9 9x^3\,dx = [x^9]_3^9 = 256 - 81 = 175$

$80 + 175 = 255$

(kanan)

$\int_1^9 9x^3\,dx = [x^9]_1^9 = 256 - 1 = 255$

[terbukti benar]

(23.) $\int_0^9 \frac{1}{\sqrt{x}}\,dx$

$\int_0^9 x^{-\frac{1}{2}}\,dx$

$\left[\frac{x^{\frac{1}{2}}}{\frac{1}{2}}\right]_0^9 = \left[2x^{\frac{1}{2}}\right]_0^9$

$= \left[2\sqrt{x}\right]_0^9$

$= 2 \cdot 2 = 9$ //

(24) a. $y = 3x - 2 \mid y = x^2$

$\cancel{V = \pi \int_1^2 ((3x-2)^2 - (x^2)^2)\,dx}$

$\cancel{V =}$

$x^2 = 3x - 2$

$x^2 - 3x + 2 = 0$

$(x-1)(x-2) = 0$

$x = 1 \rightarrow x = 2$

$V = \pi \int_1^2 ((3x-2)^2 - (x^2)^2)\,dx$

$= \pi \int_1^2 (9x^2 - 12x + 4 - x^4)\,dx$

$= \pi \left[3x^3 - 6x^2 + 4x - \frac{x^5}{5}\right]_1^2$

$= \frac{8\pi}{15}$ //

b. $y = x - 1$ & $y = 3 - x$

$x - 1 = 3 - x \rightarrow x = 2$

$V = \pi \int_1^2 (x-1)^2\,dx + \pi \int_2^3 (3-x)^2\,dx$

$V = \frac{\pi}{3} + \frac{\pi}{3}$

$V = \frac{2\pi}{3}$ //

c. $y = \sin x,\ 0 \le x \le \frac{\pi}{2}$

$V = \pi \int_0^{\frac{\pi}{2}} \sin^2 x\,dx$

$\int_0^{\pi/2} \sin^2 x\,dx = \frac{\pi}{4}$

$V = \pi \cdot \frac{\pi}{4} = \frac{\pi^2}{4}$

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(25) a. $y = \sqrt{x} \rightarrow x = y^2$, batas $0 \leq y \leq 1$

$$V = \pi \int_0^1 (y^2)^2 \, dy = \pi \int_0^1 y^4 \, dy = \frac{\pi}{5}$$

b. $y = x + 1 \rightarrow x = y - 1$, batas $1 \leq y \leq 4$

$$V = \pi \int_1^4 (y-1)^2 \, dy = \pi \left[\frac{(y-1)^3}{3} \right]_1^4 = 9\pi$$

c. $y = 9 - x^2 \rightarrow x^2 = 9 - y$, batas $-9 \leq y \leq 9$

$$V = \pi \int_{-9}^{9} (9 - y) \, dy = 162\pi$$

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